THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 12: **Tales by Dots and Lines** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The median of 37, 98, 75, 21, 17, 74, 91, 56 is: 1

(a) 17 **(b)** 58.62 **(c)** 65 **(d)** 57

2. The mean of 5 numbers is 26. The number to be added so that the mean becomes 40 is: 1

(a) 40 **(b)** 115 **(c)** 14 **(d)** 110

3. The mean of the first 13 multiples of 6 (that is 6, 12, ..., 78) is: 1

(a) 38 **(b)** 84 **(c)** 46 **(d)** 42

4. On a line graph, a town's population (in thousands) rises from 80 to 100. The percentage increase is: 1

(a) 12% **(b)** 25% **(c)** 20% **(d)** 35%

5. The mean mark of 30 students is 18 and of another 20 students is 33. The mean of all 50 students is: 1

(a) 26 **(b)** 1200 **(c)** 25.5 **(d)** 24

6. The mean of 27, 31, 44, 47, 23, 14 is 31. If 44 is replaced by 50, the new mean is: 1

(a) 37 **(b)** 31 **(c)** 39.33 **(d)** 32

7. The median of 58, 20, 78, 37, 61, 76, 49, 14 is: 1

(a) 53.5 **(b)** 61 **(c)** 50 **(d)** 49.12

8. The mean of 5 numbers is 19. The number to be added so that the mean becomes 22 is: 1

(a) 42 **(b)** 37 **(c)** 3 **(d)** 22

9. The mean of the first 7 multiples of 9 (that is 9, 18, ..., 63) is: 1

(a) 36 **(b)** 40 **(c)** 32 **(d)** 72

10. On a line graph, a town's population (in thousands) rises from 25 to 35. The percentage increase is: 1

(a) 20% **(b)** 40% **(c)** 50% **(d)** 10%

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mean of the first 9 natural numbers that are multiples of 7. 2

12. The mean marks of 25 boys is 32 and of 15 girls is 48. Find the mean mark of the whole class of 40. 2

13. The mean of 5 observations 19, 37, 23, 21, 40 is 28. Find the mean if 15 is added to every observation. 2

14. Find the mean of the first 14 natural numbers that are multiples of 2. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The mean weight of 25 students is 30 kg and of another 25 students is 34 kg. Find the mean weight of all of them. 3

16. The mean of 5 numbers is 22. If one number 13 is replaced by 23, find the new mean and the change in the mean. 3

17. The median of 30, 66, 51, 15, 22, 68, 41 is 41. If the largest value 68 is changed to 204, what happens to the median and the mean? 3

OR

The mean weight of 30 students is 28 kg and of another 20 students is 33 kg. Find the mean weight of all of them. 3

18. The mean of 5 numbers is 27. If one number 16 is replaced by 26, find the new mean and the change in the mean. 3

OR

The median of 27, 28, 30, 71, 53, 55, 59 is 53. If the largest value 71 is changed to 213, what happens to the median and the mean? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 12, 16, 23, 27, 22, 17. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 12 students has a mean mark of 70. Another 18 students with mean 80 join it. Find the mean of the combined group and the change in the original mean. 5

20. The temperatures (°C) at 8 am, 10 am, 12 noon, 2 pm, 4 pm and 6 pm were 19, 21, 30, 32, 29, 20. 5

(i) Find the rise or fall between each consecutive pair.

(ii) Find the mean temperature.

(iii) In which interval was the change largest?

OR

A class of 40 students has a mean mark of 35. Another 10 students with mean 60 join it. Find the mean of the combined group and the change in the original mean. 5

END OF QUESTION PAPER Total: 40 Marks